

77

diagnoses formalized in chapters 1–4
the complete sub-catalogue of primary headaches, 55 of them leaf codes

600

typed logic nodes
composed from 11 operator kinds present in the export

0

target verdict errors in the rule test
2 616 graph-derived cases, each checked against all 77 diagnoses

3

truth values
satisfied · not satisfied · open — missing information stays visible

Why the primary headaches

Primary headaches are the core of general practice: migraine and tension-type headache account for the large majority of all cases. At the same time, telling them apart rests on features that prose can barely keep apart — bilateral against unilateral, pressing against pulsating, aggravation by physical activity against its absence.

Three further properties add difficulty: aura criteria with their own time windows, the episodic against chronic distinction drawn over frequency thresholds, and the trigeminal autonomic cephalalgias, whose delineation depends on attack duration and accompanying signs.

FIG 1 Four chapters, four kinds of difficulty operator counts are reachable nodes per chapter in the export

CHAPTER 1 · 27 DIAGNOSES Migraine The criteria mix three kinds of connective: a counted threshold, a “two of four” selection and a nested and/or. Aura time windows are TEMPORAL nodes with bounds in hours. TEMPORAL 79 · OR 63 · KOËN 29 · QUANTIFIER 32	CHAPTER 2 · 12 DIAGNOSES Tension-type Defined largely by the absence of migraine features. The graph carries them as negated groundings and counter-attributes. Duration and the “at most one of” clause are keyword attributes — the one place where chapter 2 is checked by term, not by number. QUANTIFIER 33 · KOËN 16 · NOT 3	CHAPTER 3 · 17 DIAGNOSES Trigeminal autonomic Characteristic is the breadth of alternatives: autonomic signs enter as wide OR groups. Bout and remission patterns are carried by all 11 PERIOD_PATTERN nodes of the sub-catalogue. OR 126 · QUANTIFIER 40 · TEMPORAL 22 · PERIOD_PATTERN 11	CHAPTER 4 · 21 DIAGNOSES Other primary A heterogeneous group, often defined by a trigger — cough, exertion, cold stimulus, sexual activity. Triggers are events, not properties: all 23 causal relation nodes of the sub-catalogue sit here. Shallow trees, depth 5. TEMPORAL 31 · CAUSAL 23 · QUANTIFIER 20 · OR 16
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FIG 2 The graph of the sub-catalogue 1 348 nodes and 1 851 edges; the four chapters emerge as density zones

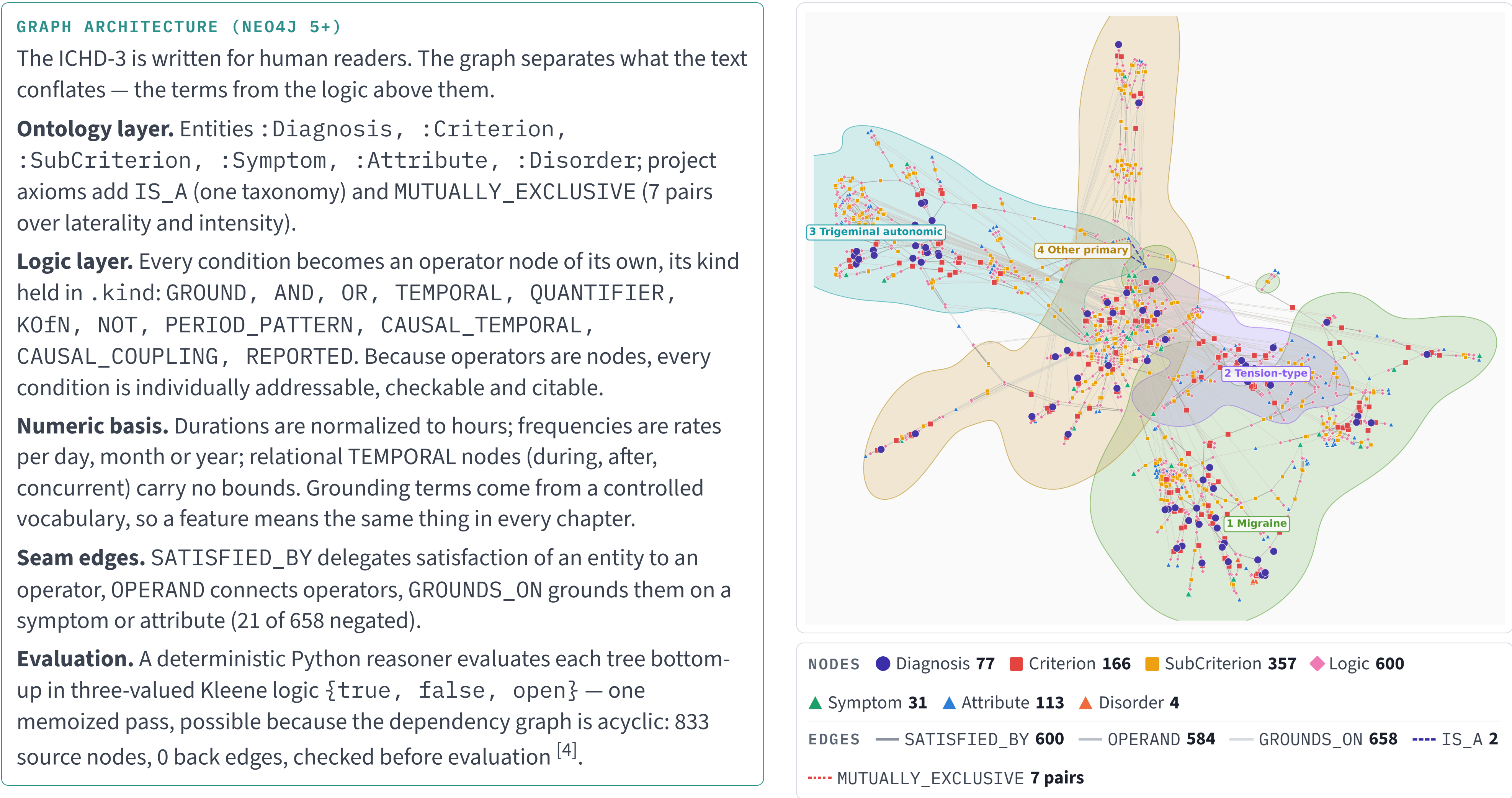


FIG 3 The same case, checked against two diagnoses verdicts as produced by the reasoner on the delivered graph

THE FACTS FROM THE NOTE Ten attacks over the past year, each lasting six to ten hours. Unilateral and pulsating. Light and noise hard to bear. Not stated: nausea, pain intensity, behaviour on exertion. HOW TO READ IT ✓ satisfied — evidenced in the note ✗ not satisfied — the note contradicts it ⚠ open — no information given Open only blocks a diagnosis that needs the branch. Absent is only what the note states as absent — the graph never infers absence from silence (open world). Criterion E (“not better accounted for by another diagnosis”) is a policy placeholder, not a node — it is left out of both trees.	1.1 Migraine without aura ✓satisfied ✓ A at least 5 attacks 10 ✓ B duration 4–72 h 6–10 h ✓ C two of four characteristics 2 ✓ unilateral location ✓ pulsating quality ⚠ moderate or severe intensity ⚠ aggravation by / avoidance of activity ✓ D at least one of two ⚠ nausea and/or vomiting ✓ photophobia and phonophobia The evidence suffices: the threshold is reached, and in D one option is enough.	2.2 Frequent episodic tension-type headache ✗not satisfied ✗ A episodes on 1–14 days a month and ≥ 12 days a year 10 / year ✓ at least 10 episodes ✗ at least 12 headache days a year ⚠ 1–14 days a month over ≥ 3 months ✓ B duration 30 min to 7 days 6 - 10 h ⚠ C two of four characteristics open ⚠ bilateral location ⚠ pressing or tightening ⚠ mild or moderate intensity ⚠ not aggravated by activity ⚠ D both conditions together open ⚠ no nausea or vomiting ⚠ at most one of photophobia / phonophobia A single evidenced no decides — here the yearly frequency — even though C and D stay open.
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Criteria given in abbreviated form; node verdicts taken from the actual reasoner run on the delivered graph.

Rule-test evaluation

788 / 788

generated cases recognized by their own diagnosis

298 / 298

refuted cases rejected

1 530 / 1 530

near-miss cases, one feature short, not accepted

Other diagnoses answering TRUE on a positive case are hierarchical parents or diagnoses that reference the case's base diagnosis by design — complications such as status migrainosus, and the probable forms. Two findings remain: the overlap of 3.1 and 3.2, which ICHD-3 separates only through criterion E, and one modelling gap where 4.6.3 lacks the exclusion of 4.6.1 and 4.6.2.

When a criterion is refuted, the case does not fall silent — it lands on the probable form of its diagnosis, in all six tested pairs.

An independent check. Because cases and rules come from the same source, the figures above test the translation, not the medicine. 918 cases coded independently by Zhang and Cheng for 16 of these diagnoses are recognized in 96 % under a closed-world reading and 34 % under an open-world one — the gap is absence information that no clinical note states.

Limitations and outlook

Sub-catalogue. Secondary headaches and neuropathies are out of scope here — they are part of the full graph.

Openness has a price. Criteria phrased as absences stay open unless a note states the absence; the closed-world reading trades that safety for recall.

Extraction. What is missed when reading the text cannot be compensated by the graph; facts and path therefore stay visible.

Course over time. Frequencies are not converted between periods; a lifetime count does not become a monthly rate.

Next: the first comparison with a purely learned approach on real outpatient histories has been run; see [2].

References

[1] ICHD-3. Cephalalgia 2018;38(1):1–211. · [2] Zwanzig D. Benchmarking Prompting Strategies for Open-Source Language Models in ICHD-3 Headache Classification. Stud Health Technol Inform 2026. doi:10.3233/SHIT260922 · [3] Zahn A, Strauß S, Zwanzig D. Towards Community-Based Evaluation of AI in Neurology. Stud Health Technol Inform 2025. doi:10.3233/SHIT251535 · [4] Kleene SC. Introduction to Metamathematics. 1952. · [5] Costabile R et al. Theory Pract Log Program 2020.

Contact

Dorian Zwanzig, M.Sc. · HNEE · dorian.zwanzig@hnee.de
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